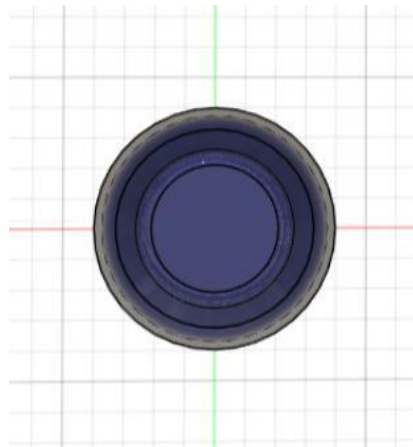


DESIGNING 3D WATER BOTTLE USING FUSION 360



DESCRIPTION

This project details the parametric 3D modeling of a 1-liter insulated water bottle using Autodesk Fusion 360. The design consists of two primary components: the bottle body with outer steel shell and inner water-filled chamber, and a threaded screw cap for secure sealing. Key features include precise volume control for the inner chamber at exactly 1 liter (1000 cm³), ergonomic shaping for grip, and a dynamic animation demonstrating cap rotation to open and close via helical motion. The model leverages Fusion 360's parametric capabilities, allowing easy scaling or modifications for manufacturing tolerances.

OUTCOMES/LEARNINGS

I got hands-on with Fusion 360's parametric tools, linking sketches to nail that exact 1L volume spot-on. Surfacing like Loft and Coil turned tricky threads and curves into something I could prototype confidently. Animation was a game-changer—Motion Link synced the cap's twist and slide so smoothly it felt real. Debugging joint glitches sharpened my assembly skills for foolproof motion. Rendering steel took a few tries but now pops for presentations. Section Analysis is my new quick-check essential for

volumes. This sped my full workflow from idea to animated demo, prepping me for deadline-driven engineering work.